ECx00x&EG800K&EG810M&EG915N&EG950A Series GNSS Application Note

LTE Standard Module Series

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About the Document

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1 Introduction

Quectel ECx00x-CN, EG800K-CN, EG810M-CN, EG915N series and EG950A-EL modules integrate the high-performance multi-GNSS engine supporting multiple constellations. The multi-GNSS engine is suitable for various application scenarios with low cost and accurate positioning requirements, such as turn-by-turn navigation, asset tracking, personnel tracking, location-aware games, as well as home and fleet management.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
ECx00x-CN	EC200A-CN
	EC800M-CN
-	EG800K-CN
-	EG810M-CN
-	EG915N Series
-	EG950A-EL

NOTE

Among the above applicable modules, the ECx00x-CN, EG810M-CN, EG915N series and EG950A-EL modules are available with optional GNSS functionality. Consult Quectel Technical Support to find out if your module model supports GNSS.

1.2. Enable/Disable GNSS Function

The module supports location calculation without any network assistance. To enable/disable the GNSS feature follow the steps below:

Step 1: Configure GNSS parameters with **AT+QGPSCFG**. This step can be omitted if default configurations are used.

Step 2: Enable the GNSS feature with **AT+QGPS=1**.

Step 3: Once position is fixed successfully, obtain the positioning information by using any of the following methods:

(1) NMEA sentences are output through USB NMEA port by default. Or you can set another port as the NMEA sentence output port with **AT+QGPSCFG="outport"**.

(2) Obtain positioning information such as latitude, longitude, altitude, GNSS positioning mode, time, and number of satellites used for location calculation, directly with **AT+QGPSLOC**.

(3) You can also obtain positioning information via the AT port. To enable acquisition of specified NMEA sentences with **AT+QGPSGNMEA** via AT port, execute **AT+QGPSCFG="nmeasrc",1**. If **<NMEA_src>** is set to 0 with **AT+QGPSCFG="nmeasrc",0** the specified NMEA sentences cannot be acquired with **AT+QGPSGNMEA**.

Step 4: Disable the GNSS feature with **AT+QGPS=0** or **AT+QGPSEND**.

1.3. Supported NMEA Sentence Types

The default NMEA sentences supported by the module are compatible with NMEA 0183 protocol. NMEA sentences of different satellite systems are differentiated by the following prefixes:

GPS NMEA sentences have the prefix "GP":

- GPGGA – Global positioning system fix data, such as time and position.
- GPRMC – Recommended minimum specific GNSS data.
- GPGSV – GPS satellites in view (number of satellites in view, satellite ID numbers etc.).
- GPGSA – GNSS DOP and active satellites.
- GPVTG – Course over ground and ground speed.
- GPGLL – Geographic position – latitude/longitude.
- GPGST – GNSS pseudorange error statistics.

BDS NMEA sentences have the prefix "BD":

- BDGGA – Global positioning system fix data, such as time and position.
- BDRMC – Recommended minimum specific GNSS data.
- BDGSV – BDS satellites in view (number of satellites in view, and satellite ID numbers, etc.).

- BDGSA – GNSS DOP and active satellites.
- BDVTG – Course over ground and ground speed.
- BDGLL – Geographic position – latitude/longitude.
- BDGST – GNSS pseudorange error statistics.

Galileo NMEA sentences have the prefix "GA":

- GAGGA – Global positioning system fix data, such as time and position.
- GARMC – Recommended minimum specific GNSS data.
- GAGSV – Galileo satellites in view (number of satellites in view, and satellite ID numbers, etc.).
- GAGSA – GNSS DOP and active satellites.

GLONASS NMEA sentences have the prefix "GL":

- GLGGA – Global positioning system fix data, such as time and position.
- GLRMC – Recommended minimum specific GNSS data.
- GLGSV – GLONASS satellites in view (number of satellites in view, and satellite ID numbers etc.).
- GLGSA – GNSS DOP and active satellites.

GNSS multi-constellation NMEA sentences have the prefix “GN”:

- GNGGA – Global positioning system fix data, such as time and position.
- GNRMC – Recommended minimum specific GNSS data.
- GNGSA – GNSS DOP and active satellites.
- GNVTG – Course over ground and ground speed.
- GNGLL – Geographic position – latitude/longitude.
- GNGST – GNSS pseudorange error statistics.

NOTE

1. The QZSS enable status aligns with that of the GPS system.
2. Support for Galileo, GLONASS and QZSS satellite systems varies depending on module model. Contact Quectel Technical Support for details.

2 Description of GNSS AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Types of AT Commands

Command Type	Syntax	Description
Test Command	**AT+<cmd>=?**	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	**AT+<cmd>?**	Check the current parameter value of the corresponding command.
Write Command	**AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]**	Set user-definable parameter value.
Execution Command	**AT+<cmd>**	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence. The URLs, domain names, IP addresses, usernames/accounts, and passwords (if any) in the AT command examples are provided for illustrative and explanatory purposes only, and they should be modified to reflect your actual usage and specific needs.

2.3. AT Command Description

2.3.1. AT+QGPSCFG Configure GNSS Settings

This command queries and configures various GNSS settings, including the output port and output types of NMEA sentences.

AT+QGPSCFG Configure GNSS Settings	
Test Command **AT+QGPSCFG=?**	Response **+QGPSCFG: "outport",(**list of supported **<out_port>**s**)** **+QGPSCFG: "nmeasrc",(**list of supported **<NMEA_src>**s**)** **+QGPSCFG: "gpsnmeatype",(**list of supported **<GPS_NMEA_type>**s**)** **+QGPSCFG: "gnssconfig",(**list of supported **<GNSS_config>**s**)** **+QGPSCFG: "autogps",(**list of supported **<autoGPS>**s**)** **+QGPSCFG: "apflash",(**list of supported **<AP_flash_mode>**s**)** **+QGPSCFG: "urc",<bitmap>** **+QGPSCFG: "ntp",<NTP_server>** **OK**

2.3.1.1. AT+QGPSCFG="outport" Configure Output Port of NMEA Sentences

This command queries and configures the output port of NMEA sentences.

AT+QGPSCFG="outport" Configure Output Port of NMEA Sentences	
Write Command **AT+QGPSCFG="outport"[,<out_port**	Response If the optional parameter is omitted, query the current setting:

>]	**+QGPSCFG: "outport",<out_port>** **OK** If the optional parameter is specified, set the output port of NMEA sentences: **OK** Or **ERROR** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved to NVRAM automatically.

Parameter

<out_port>	String type. Output port of NMEA sentences. "none" Close NMEA sentence output port "usbnmea" Output through USB NMEA port "uartdebug" Output through debug UART port
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

NOTE

NMEA sentences are output through USB NMEA port by default when USB NMEA port is enabled. If the USB NMEA port is not enabled, NMEA sentences are not output by default. To output NMEA sentences through the debug UART port, you must configure it manually with **AT+QGPSCFG="outport",<out_port>**.

2.3.1.2. AT+QGPSCFG="nmeasrc" Enable/Disable Acquisition of NMEA Sentences with AT+QGPSGNMEA

This command enables or disables acquisition of NMEA sentences with **AT+QGPSGNMEA**.

AT+QGPSCFG="nmeasrc" Enable/Disable Acquisition of NMEA Sentences with AT+QGPSGNMEA	
Write Command **AT+QGPSCFG="nmeasrc"[,<NMEA_**	Response If the optional parameter is omitted, query the current setting:

src>]	**+QGPSCFG: "nmeasrc",<NMEA_src>** **OK** If the optional parameter is specified, enable or disable acquisition of NMEA sentences with **AT+QGPSGNMEA**: **OK** Or **ERROR** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved to NVRAM automatically.

Parameter

<NMEA_src>	Integer type. Enable or disable the acquisition of NMEA sentences with **AT+QGPSGNMEA**. If enabled, after **AT+QGPSGNMEA** is executed, NMEA sentences are output through the AT port as a return value. 0 Disable 1 Enable
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

2.3.1.3. AT+QGPSCFG="gpsnmeatype" Configure NMEA Sentence Output Type

This command configures the NMEA sentence output type.

AT+QGPSCFG="gpsnmeatype"	**Configure NMEA Sentence Output Type**
Write Command **AT+QGPSCFG="gpsnmeatype"[,<GPS_NMEA_type>]**	Response If the optional parameter is omitted, query the current setting: **+QGPSCFG: "gpsnmeatype",<GPS_NMEA_type>** **OK** If the optional parameter is specified, set the NMEA sentence output type: **OK** Or **ERROR**

	If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved to NVRAM automatically.

Parameter

<GPS_NMEA_type>	Integer type. Output type of NMEA sentences by a XOR operation. Range: 0–127. 0 Disabled 1 GGA 2 RMC 4 GSV 8 GSA 16 VTG 31 GGA, RMC, GSV, GSA and VTG 32 GLL 63 GST 127 All the above types of sentences
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

NOTE

1. This command can be executed to configure the output type of NMEA sentences even if the GNSS feature is disabled. The configuration takes effect after the GNSS feature is enabled.
2. Some module models do not support GST sentence. Contact Quectel Technical Support for details.

2.3.1.4. AT+QGPSCFG="gnssconfig" Configure supported GNSS Constellations

This command configures the supported GNSS constellations of the module.

AT+QGPSCFG="gnssconfig" Configure Supported GNSS Constellations	
Write Command **AT+QGPSCFG="gnssconfig"[,<GNSS_config>]**	Response If the optional parameter is omitted, query the current setting: **+QGPSCFG: "gnssconfig",<GNSS_config>** **OK** If the optional parameter is specified, set the supported

	GNSS constellations: **OK** Or **ERROR** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved to NVRAM automatically.

Parameter

<GNSS_config>	Integer type. Enabled GNSS constellations. Different module models support different constellations. If **<GNSS_config>** is set to the value which is not supported by the module, the command does not take effect. 0 GPS 1 GPS + BDS 3 GPS + GLONASS + Galileo 4 GPS + GLONASS 5 GPS + BDS + Galileo 6 GPS + Galileo 7 BDS 8 GPS + BDS + GLONASS
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

NOTE

1. When the GNSS feature is not enabled and configured, the **<GNSS_config>** read by this command is 0, which has no relevance.
2. The default constellation configuration of this command supports GPS + BDS + Galileo. If the module does not support Galileo, the default configuration is GPS + BDS.
3. For details on the GNSS constellation(s) supported by different module models, contact Quectel Technical Support.

2.3.1.5. AT+QGPSCFG="autogps" Enable/Disable Automatic Running of GNSS

This command enables or disables the automatic running of GNSS after the module is powered on.

AT+QGPSCFG="autogps" Enable/Disable Automatic Running of GNSS	
Write Command **AT+QGPSCFG="autogps"[,<autoGPS>]**	Response If the optional parameter is omitted, query the current setting: **+QGPSCFG: "autogps",<autoGPS>** **OK** If the optional parameter is specified, enable or disable the automatic running of GNSS: **OK** Or **ERROR** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module is rebooted. The configuration is saved to NVRAM automatically.

Parameter

<autoGPS>	Integer type. Enable or disable automatic running of GNSS. 0 Disable 1 Enable
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

2.3.1.6. AT+QGPSCFG="apflash" Enable/Disable AP-Flash Quick Hot Start

This command enables or disables the AP-Flash quick hot start solution. After the GNSS feature is enabled, the positioning time can be reduced by importing ephemeris data into the GNSS module upon its next startup. (The ephemeris data is valid for 1 hour and is automatically updated every hour).

AT+QGPSCFG="apflash" Enable/Disable AP-Flash Quick Hot Start	
Write Command **AT+QGPSCFG="apflash"[,<AP_flash_mode>]**	Response If the optional parameter is omitted, query the current setting: **+QGPSCFG: "apflash",<AP_flash_mode>**

	OK If the optional parameter is specified, enable or disable the AP-Flash quick hot start solution: **OK** Or **ERROR** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved to NVRAM automatically.

Parameter

<AP_flash_mode>	Integer type. Enable or disable the AP-Flash quick hot start solution. 0 Disable 1 Enable
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

2.3.1.7. AT+QGPSCFG="urc" Configure Reporting Type of GNSS URC

This command configures the reporting type of GNSS URC.

AT+QGPSCFG="urc" Configure Reporting Type of GNSS URC	
Write Command **AT+QGPSCFG="urc"[,<bitmap>]**	Response If the optional parameter is omitted, query the current setting: **+QGPSCFG: "urc",<bitmap>** **OK** If the optional parameter is specified, set the reporting type of GNSS URC: **OK** Or **ERROR** If there is any error related to ME functionality: **+CME ERROR: <errcode>**

Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved to NVRAM automatically.

Parameter

<bitmap>	Hexadecimal type. Bit mask of GNSS URC reporting type. Example: 0x5 = 0x1 (report GNSS firmware loading URC) + 0x4 (report AGNSS URC) 0x0 Disable GNSS URC reporting 0x1 Enable reporting GNSS firmware loading URC **+QGPS: "firmware",<reserved>,<firmware_code>** 0x4 Enable reporting AGNSS URC **+QGPS: "agps",<reserved>,<AGPS_code>** 0x5 Enable reporting GNSS firmware loading and AGNSS URCs 0xFF Enable reporting all GNSS URCs
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.
<reserved>	Integer type. Reserved. Fixed at 0.
<firmware_code>	Integer type. Firmware loading status. 0 Successful firmware loading. 1 Failed firmware loading.
<AGPS_code>	Integer type. AGNSS status. 0 Successful AGNSS file downloading and application. 5 Dial-up activation error. 6 Socket connection error.

2.3.1.8. AT+QGPSCFG="ntp" Configure GNSS NTP Server Address

This command configures the GNSS NTP server address.

AT+QGPSCFG="ntp" Configure GNSS NTP Server Address	
Write Command **AT+QGPSCFG="ntp"[,<NTP_server>]**	Response If the optional parameter is omitted, query the current setting: **+QGPSCFG: "ntp",<NTP_server>** **OK** If the optional parameter is specified, set the GNSS NTP server address. **OK** Or

	ERROR If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved to NVRAM automatically.

Parameter

<NTP_server>	String type. GNSS NTP server address. The default value can be obtained by executing **AT+QGPSCFG="ntp"** before the initial configuration.
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

2.3.2. AT+QGPS Enable/Disable GNSS Feature

This command enables or disables the GNSS feature.

AT+QGPS Enable/Disable GNSS Feature	
Test Command **AT+QGPS=?**	Response **+QGPS: (**list of supported **<GNSS_state>**s**)** **OK**
Read Command **AT+QGPS?**	Response **+QGPS: <GNSS_state>** **OK**
Write Command **AT+QGPS=<GNSS_state>**	Response **OK** Or **ERROR** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<GNSS_state>	Integer type. Enable/disable GNSS feature. 0 Disable GNSS 1 Enable GNSS
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

2.3.3. AT+QGPSEND Disable GNSS Feature

This command disables GNSS feature.

AT+QGPSEND Disable GNSS Feature	
Test Command **AT+QGPSEND=?**	Response **OK** Or **ERROR**
Read Command **AT+QGPSEND?**	Response **OK** Or **ERROR**
Execution Command **AT+QGPSEND**	Response **OK** Or **ERROR** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	/

Parameter

<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

2.3.4. AT+QGPSLOC Acquire Positioning Information

This command acquires positioning information. Before executing this command, GNSS function should be enabled with **AT+QGPS**. If the position is not fixed, **+CME ERROR: <errcode>** is returned to indicate the corresponding issue.

AT+QGPSLOC Acquire Positioning Information	
Test Command **AT+QGPSLOC=?**	Response **+QGPSLOC: <UTC>,<latitude>,<longitude>,<HDOP>,<altitude>,(**list of supported **<fix>**s**),<COG>,<spkm>,<spkn>,<date>,<nsat>** **OK**
Write Command **AT+QGPSLOC=<mode>**	Response **+QGPSLOC: <UTC>,<latitude>,<longitude>,<HDOP>,<altitude>,<fix>,<COG>,<spkm>,<spkn>,<date>,<nsat>** **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	/

Parameter

<mode>	Integer type. Latitude and longitude display format. 0 **<latitude>,<longitude>** format: ddmm.mmmmN/S,dddmm.mmmmE/W 1 **<latitude>,<longitude>** format: ddmm.mmmmmm,N/S,dddmm.mmmmmm,E/W 2 **<latitude>,<longitude>** format: (-)dd.ddddd,(-)ddd.ddddd
<UTC>	String type (without double quotation marks). UTC time. Format: hhmmss.ss. (Quoted from GNRMC sentence.)
<latitude>	String type (without double quotation marks). Latitude. If **<mode>** is 0: Format: ddmm.mmmmN/S (Quoted from GNRMC sentence.) dd Degree. Range: 00–89. mm.mmmm Minute. Range: 00.0000–59.9999. N/S North latitude/South latitude. If **<mode>** is 1: Format: ddmm.mmmmmm,N/S. dd Degree. Range: 00–89. mm.mmmmmm Minute. Range: 00.000000–59.999999. N/S North latitude/South latitude. If **<mode>** is 2: Format: (-)dd.ddddd.

	dd.ddddd Degree. Range: -89.99999 to 89.99999. - South latitude.
<longitude>	String type (without double quotation marks). Longitude. If **<mode>** is 0: Format: dddmm.mmmmE/W (Quoted from GNRMC sentence.) ddd Degree. Range: 000–179. mm.mmmm Minute. Range: 00.0000–59.9999. E/W East longitude/West longitude. If **<mode>** is 1: Format: dddmm.mmmmmm,E/W (Quoted from GNRMC sentence.) ddd Degree. Range: 000–179. mm.mmmmmm Minute. Range: 00.000000–59.999999. E/W East longitude/West longitude. If **<mode>** is 2: Format: (-)ddd.ddddd (Quoted from GNRMC sentence.) ddd.ddddd Degree. Range: -179.99999 to 179.99999. - West longitude
<HDOP>	Double type. Horizontal dilution of precision. Accurate to one decimal place. (Quoted from GPGSA sentence.)
<altitude>	Double type. Altitude of the antenna above the sea level. Unit: meter. Accurate to one decimal place. (Quoted from GNGGA sentence.)
<fix>	Integer type. GNSS positioning mode. (Quoted from GPGSA sentence.) 1 No positioning 2 2D positioning 3 3D positioning
<COG>	String type (without double quotation marks). Course Over Ground based on true north. The output is null if the speed measurement fails or the speed is extremely low in a static scenario. (Quoted from GNRMC sentence.)
<spkm>	Double type. Speed over ground. Unit: km/h. Accurate to one decimal place. (Quoted from GNRMC sentence.)
<spkn>	Double type. Speed over ground. Unit: knots. Accurate to one decimal place. (Quoted from GNRMC sentence.)
<date>	String type (without double quotation marks). UTC date. Format: ddmmyy. (Quoted from GNRMC sentence.) dd Day mm Month yy Year
<nsat>	Integer type. Number of satellites used for location calculation. It is in two-digit number format, where single-digit numbers are prefixed with a “0”. (Quoted from GNGGA sentence.)
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

2.3.5. AT+QGPSGNMEA Acquire Specified NMEA Sentences

This command acquires specified NMEA sentences. Before using this command, enable GNSS with **AT+QGPS**, and set **<NMEA_src>** to 1 to enable acquisition of NMEA sentences with **AT+QGPSGNMEA**.

The sentence output can be disabled with **AT+QGPSCFG="gpsnmeatype",0**. This command prevents the module from outputting updated sentences. If you disable sentence output after acquiring NMEA sentences, the module will save the sentences obtained between the time GNSS was enabled and when sentence output was disabled. At this time, you can still retrieve the saved NMEA sentence with **AT+QGPSGNMEA**, as long as it matches the specified sentence type.

AT+QGPSGNMEA Acquire Specified NMEA Sentences	
Test Command **AT+QGPSGNMEA=?**	Response **+QGPSGNMEA: (**list of supported **<NMEA_type>**s**)** **OK**
Write Command Query GGA sentence **AT+QGPSGNMEA="GGA"**	Response **[+QGPSGNMEA: <GGA_sentence>]** **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Write Command Query RMC sentence **AT+QGPSGNMEA="RMC"**	Response **[+QGPSGNMEA: <RMC_sentence>]** **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Write Command Query GSV sentence **AT+QGPSGNMEA="GSV"**	Response **[+QGPSGNMEA: <GSV_sentence>]** **[…]** **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Write Command Query GSA sentence **AT+QGPSGNMEA="GSA"**	Response **[+QGPSGNMEA: <GSA_sentence>]** **[…]** **OK**

	If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Write Command Query VTG sentence **AT+QGPSGNMEA="VTG"**	Response **[+QGPSGNMEA: <VTG_sentence>]** **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Write Command Query GLL sentence **AT+QGPSGNMEA="GLL"**	Response **[+QGPSGNMEA: <GLL_sentence>]** **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Write Command Query GST sentence **AT+QGPSGNMEA="GST"**	Response **[+QGPSGNMEA: <GST_sentence>]** **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	/

Parameter

<NMEA_type>	String type. NMEA sentence type. "GGA" GGA sentence "RMC" RMC sentence "GSV" GSV sentence "GSA" GSA sentence "VTG" VTG sentence "GLL" GLL sentence "GST" GST sentence
<GGA_sentence>	String type (without double quotation marks). GGA sentences.
<RMC_sentence>	String type (without double quotation marks). RMC sentences.
<GSV_sentence>	String type (without double quotation marks). GSV sentences.
<GSA_sentence>	String type (without double quotation marks). GSA sentences.

<VTG_sentence>	String type (without double quotation marks). VTG sentences.
<GLL_sentence>	String type (without double quotation marks). GLL sentences.
<GST_sentence>	String type (without double quotation marks). GST sentences.
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

2.3.6. AT+QAGPS Enable/Disable AGNSS

This command enables or disables AGNSS function of GNSS.

AT+QAGPS Enable/Disable AGNSS	
Test Command **AT+QAGPS=?**	Response **+QAGPS: (**list of supported **<enable>**s**)** **OK**
Read Command **AT+QAGPS?**	Response **+QAGPS: <enable>** **OK**
Write Command **AT+QAGPS=<enable>**	Response **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module is rebooted. The configuration is saved to NVRAM automatically.

Parameter

<enable>	Integer type. Enable or disable AGNSS function of GNSS. 0 Disable 1 Enable
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

NOTE

1. One PDP context should be activated before this command is executed to enable AGNSS, otherwise the AGNSS function will not take effect.
2. The use of the AGNSS function will consume some data traffic .

2.3.7. AT+QAGPSCFG Configure AGNSS Parameters

When the AGNSS server information is changed, this command configures AGNSS parameters.

AT+QAGPSCFG Configure AGNSS Parameters	
Test Command **AT+QAGPSCFG=?**	Response **+QAGPSCFG: (**list of supported **<profile>**s**),<URL>,<vendorID>,<modelID>,<password>** **OK**
Read Command **AT+QAGPSCFG?**	Response **+QAGPSCFG: <profile>,<URL>,<vendorID>,<modelID>,<password>** **OK**
Write Command **AT+QAGPSCFG=<profile>[,<URL>[,<vendorID>[,<modelID>[,<password>]]]]**	Response **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module is rebooted. The configuration is saved to NVRAM automatically.

Parameter

<profile>	Integer type. PDP index. Range: 1–7. Default value: 1.
<URL>	String type. AGNSS server address.
<vendorID>	String type. Username. Maximum length: 30 bytes.
<modelID>	String type. Customer identification.
<password>	String type. Password. Maximum length: 30 bytes.
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

NOTE

1. **<profile>** is reserved.
2. This command can only be used to configure AGNSS parameters when the AGNSS server information has been changed. Before configuring AGNSS parameters, you can query the default values of the optional parameters by executing **AT+QAGPSCFG?**.

2.3.8. AT+QGPSVBCKP Configure GNSS Backup Power Supply

This command configures the GNSS backup power supply function.

AT+QGPSVBCKP Configure GNSS Backup Power Supply	
Test Command **AT+QGPSVBCKP=?**	Response **+QGPSVBCKP: (**list of supported **<enable>**s**)** **OK**
Read Command **AT+QGPSVBCKP?**	Response **+QGPSVBCKP: <enable>** **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Write Command **AT+QGPSVBCKP=<enable>**	Response **OK** If there is any error related to ME functionality: **+CME ERROR: <errcode>**
Maximum Response Time	300 ms
Characteristics	For the configuration to take effect, the command must be set before GNSS is enabled. The configuration is not saved.

Parameter

<enable>	Integer type. Whether to connect the backup power supply pin on the GNSS hardware. 0 Not connect 1 Connect
<errcode>	Integer type. Error code. See ***Chapter 4*** for details.

NOTE

1. The configuration is not saved on power loss. This command must be executed before GNSS is enabled, otherwise the command will not take effect.
2. Only some module models support this command. Contact Quectel Technical Support for details.

3 Example

3.1. Set USB NMEA Port as NMEA Sentence Output Port

Set the output port of NMEA sentences to USB NMEA port with **AT+QGPSCFG="outport","usbnmea"**, as shown below:

```
AT+QGPSCFG="outport","usbnmea"  //Set the output port of NMEA sentences to USB NMEA port.
OK
AT+QGPS=1                       //Enable GNSS feature.
OK
```

3.2. Enable/Disable GNSS Feature

In this example, default parameters are used to enable the GNSS feature. After GNSS is enabled, it can be disabled with **AT+QGPSEND**.

```
AT+QGPS=1                       //Enable GNSS feature.
OK
AT+QGPSLOC=0                    //Acquire positioning information.
+QGPSLOC: 063416.40,3143.2951N,11713.0655E,0.6,224.9,2,162.57,17.6,9.5,110620,07

OK
AT+QGPSEND                      //Disable GNSS feature.
OK
```

3.3. Acquire Specified NMEA Sentences Using <NMEA_src>

When the GNSS feature is enabled and **<NMEA_src>** is set to 1, NMEA sentences can be acquired directly with **AT+QGPSGNMEA**.

```
AT+QGPSCFG="nmeasrc",1          //Set <NMEA_src> to 1 to enable acquisition of NMEA
OK                              sentences with AT+QGPSGNMEA.
```

```
AT+QGPSGNMEA="GGA"                  //Acquire GGA sentence.
+QGPSGNMEA: $GPGGA,103647.000,3150.721154,N,11711.925873,E,1,02,4.7,59.8,M,-2.0,M,,*77

OK
AT+QGPSCFG="nmeasrc",0              //Set <NMEA_src> to 0 to disable acquisition of NMEA
OK                                  sentences with AT+QGPSGNMEA.
AT+QGPSGNMEA="GGA"                  //Acquire GGA sentence.
+CME ERROR: 507                     //Acquisition of NMEA sentences through
                                    AT+QGPSGNMEA was disabled, and thus GGA
                                    sentences cannot be acquired.
```

3.4. Enable AGNSS Function

Ephemeris data can be acquired automatically every time the module is powered on and the GNSS is enabled, provided that the network is normal, to achieve quick positioning.

```
AT+QIACT=1                          //Activate PDP context 1.
OK
AT+QAGPS=1                          //Enable AGNSS function.
OK
AT+QGPS=1                           //Enable GNSS feature.
OK
```

NOTE

AT+QIACT is used for reference only. You can activate any PDP context in practical applications. For details on **AT+QIACT**, see Quectel_EC200x&EG800K&EG91xN&EG912Y&EG950A_Series_TCP(IP)_Application_Note.

3.5. Report Firmware Loading URC

```
AT+QGPSCFG="urc"                    //Query the current setting.
+QGPSCFG: "urc",0x1                 //The reporting of firmware loading URC is enabled by default.

OK
AT+QGPS=1                           //Enable GNSS feature.
OK

+QGPS: "firmware",0,0               //Successful firmware loading.
```

3.6. Report AGNSS URC

```
AT+QIACT=1                  //Activate PDP context 1.
OK
AT+QAGPS=1                  //Enable AGNSS function.
OK
AT+QGPSCFG="urc",0x4        //Enable reporting AGNSS URC.
OK
AT+QGPS=1                   //Enable GNSS feature.
OK

+QGPS: "agps",0,0           //Successful AGNSS file downloading and application.
AT+QGPSEND                  //Disable GNSS feature.
OK
```

4 Summary of Error Codes

<errcode> indicates an error related to GNSS operation. The details about **<errcode>** are described in the following table.

Table 3: Summary of Error Codes

<errcode>	Description
501	Invalid parameter(s)
502	Operation not supported
503	GNSS subsystem busy
504	Session is ongoing
505	Session not active
506	Operation timeout
507	Function not enabled
508	Time information error
512	Validity time is out of range
513	Internal resource error
514	GNSS locked
516	Not fixed currently
517	CMUX port is not open
549	Unknown error

5 Appendix References

Table 4: Terms and Abbreviations

Abbreviation	Description
AGNSS	Assisted GNSS (Global Navigation Satellite System)
BDS	BeiDou Navigation Satellite System
CMUX	Connection Multiplexing
DOP	Dilution of Precision
Galileo	Galileo Satellite Navigation System (EU)
GGA	Global Positioning System Fix Data
GLL	Geographic Position – Latitude and Longitude
GLONASS	Global Navigation Satellite System (Russia)
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GSA	GPS DOP and Active Satellites
GST	GNSS Pseudorange Error Statistics
GSV	GNSS Satellites in View
ME	Mobile Equipment
NMEA	NMEA (National Marine Electronics Association) 0183 Interface Standard
NTP	Network Time Protocol
NVRAM	Non-Volatile Random Access Memory
PDP	Packet Data Protocol
RAM	Random Access Memory

RMC	Recommended Minimum Specific GNSS Data
UART	Universal Asynchronous Receiver/Transmitter
USB	Universal Serial Bus
UTC	Coordinated Universal Time
VTG	Course Over Ground & Ground Speed
XOR	Exclusive-OR (gate)